Guilherme Teixeira de Mello

Senior iOS Developer
Swift - SwiftUI - Swift concurrency - Offline-first architecture - MVVM - Clean architecture - On-device AI - CI/CD
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Portfolio website - linkedin.com/in/guilherme-demello - github.com/guilhermemello07 - FoodLens on App Store

PROFILE

Senior iOS Engineer with 6+ years shipping and modernising native iOS applications end to end. Sole iOS engineer at EIDA Solutions, owning architecture, the offline-first data layer, API integration, testing, CI/CD and App Store releases for a field application used daily by construction engineers in low-connectivity environments. Migrated a legacy RxSwift/UIKit codebase to SwiftUI, Combine (reactive and event-driven architecture) and structured concurrency, and ship production AI features built on Apple's Foundation Models framework and the Anthropic Claude API. Completing a B.Sc. in Computer Science at ATU Donegal, Ireland.

PROFESSIONAL EXPERIENCE

Senior iOS Engineer - EIDA Solutions - Dublin, Ireland / Madrid, Spain - (August 2024 - Present)
Sole iOS engineer for a native iPad / iPhone application supporting complex construction workflows used by field engineers across multiple sites, working alongside Android engineers, QA and product.

- Designed and own the offline-first data layer as an event-driven sync pipeline: local mutations are captured as durable, ordered events in SQLite/GRDB and replayed against the API on reconnect, with different conflict-resolution strategies resolving concurrent edits to the same drawing - an eventually consistent model that keeps the app fully writable on sites with no connectivity.
- Diagnosed and resolved cross-platform sync and correctness defects, including missing pagination parameters that silently truncated punch-list synchronisation, and aligned multi-signer completion gating and annotation date handling between iOS and Android.
- Led modernisation from RxSwift/UIKit to SwiftUI and structured concurrency - retiring the reactive layer in favour of Combine, async/await and Observation, with actor isolation and Sendable conformance across module boundaries.
- Architected secure authentication with Auth0 (OIDC/OAuth 2.0), Keychain-backed token storage and full session lifecycle management, with zero authentication-related production incidents.
- Own CI/CD and release end to end: XCTest and XCUITest suites, SwiftLint enforcement, and automated pipelines on GitHub Actions, Fastlane and Xcode Cloud with TestFlight delivery, sustaining a 99% crash-free session rate tracked through Firebase Crashlytics.
- Contribute directly to the Android codebase (Kotlin / Jetpack Compose) where feature parity requires it, and drive changes through structured code-review cycles on both platforms.

Selected project: Slope Verification (2026)
- Delivered a "smart annotation" system placing automated, position-aware annotations on construction drawings, replacing error-prone manual placement with algorithmically precise positioning.
- Scoped the feature through requirements interviews conducted directly with the client, deferring zoom and annotation styling to a later phase by agreement in order to protect a fixed deadline.
- Implemented low-level PDF and image manipulation with the PDFTron/Apryse SDK, including coordinate-system conversion across heterogeneous document formats.
- Shipped 20% ahead of schedule, with the technical debt accepted under deadline pressure documented at the time and cleared in a subsequent cycle. (Trade-offs and technical debt management)

iOS Developer - GEM TechLabs AI - Own Company - Part-time (Side projects) - (January 2022 - Present)
- Ship native iOS applications end to end - discovery, architecture, App Store release and post-launch iteration - using Swift, SwiftUI, Swift Concurrency and MVVM-Coordinator.
- Architected a layered on-device/cloud inference pipeline for FoodLens: Apple's Foundation Models framework within the on-device context budget, routing to the Anthropic Claude API when needed, with availability checks and graceful degradation when Apple Intelligence is unavailable.
- Applied guided generation (@Generable) to constrain model output to typed Swift structures, removing a class of parsing failures from free-form responses.
- Adopted Apple's Evaluations framework to grade AI output against defined metrics and model judges, treating non-deterministic features as a measurable testing problem rather than relying on manual spot-checks.

- Exposed core flows through App Intents for Shortcuts, Spotlight and Siri, extending the app beyond its own UI. (This helps create a new discovery surface for the app - great for consumer-facing apps and also to adopt Apple Intelligence when available)
- Use Claude Code and OpenAI Codex with custom project-level agent configuration (CLAUDE.md, AGENTS.md, SKILLS files) to reduce boilerplate while maintaining rigorous code-review standards.

iOS Developer - Cabo Frio City Hall - Brazil - (May 2021 - June 2023)
- Delivered internal iOS applications integrating municipal workflows across finance, healthcare and social-assistance departments, translating complex government processes into intuitive mobile experiences.
- Implemented secure data-transfer and authentication mechanisms to protect sensitive citizen data during inter-departmental integration, ensuring compliance with data-protection regulations.
- Served concurrently as President of the Bidding Committee, overseeing public procurement processes.

iOS Developer - MM Trade Group - Brazil (remote) - (January 2020 - May 2021)
- Developed and shipped two native iOS applications end to end using Swift and UIKit, owning the full lifecycle from concept through App Store deployment.
- Implemented RESTful API integrations and ensured adherence to Apple's Human Interface Guidelines across all interfaces.

TECHNICAL SKILLS
Languages & UI: Swift, Objective-C, SwiftUI, UIKit, App Intents, Combine, RxSwift
Concurrency: Swift 6 language mode, strict concurrency, async/await, actors, Sendable, structured concurrency, Observation, AsyncSequence
Architecture: MVVM, MVVM-Coordinator, Clean Architecture, SOLID, dependency injection, protocol-oriented programming, Swift Package Manager (SPM) modularisation
Data & Sync: Offline-first and event-driven architecture, SQLite/GRDB, Core Data, SwiftData, CloudKit, background sync queues, conflict resolution, pagination, caching
AI on Apple platforms: Foundation Models framework (on-device and Private Cloud Compute), guided generation (@Generable), tool calling, Evaluations framework, Anthropic Claude API, Open Food Facts API
Networking & Security: URLSession, Alamofire, REST, Codable, retry/backoff, Keychain, Auth0 (OIDC/OAuth 2.0), certificate/SSL pinning, LocalAuthentication, data protection
Testing & CI/CD: XCTest, XCUITest, snapshot testing, SwiftLint, Fastlane, GitHub Actions, Xcode Cloud, TestFlight, GitFlow
Other: StoreKit 2, RevenueCat, Firebase (Crashlytics, Analytics, Remote Config), PDFTron/Apryse, localisation (i18n), accessibility (VoiceOver, Dynamic Type), AI-augmented development (Claude Code, OpenAI Codex)

EDUCATION
B.Sc. Computer Science - Atlantic Technological University - Donegal (ATU - Ireland) - (Expected 2027)
Associate Degree in Computer Science - University of the People - (2018 - 2021)

CERTIFICATIONS & OTHER
CS50: Introduction to Computer Science - Harvard / edX
Google Cybersecurity Professional Certificate - Google
Anthropic Claude - Anthropic Skill Jar
Invited to give a talk at iOSDevUK 2026
Invited to WWDC25 at Apple Park
Wrote a book teaching native iOS development in Portuguese (PT-BR)

LANGUAGES
English - C2 (Fluent)
Portuguese - C2 (Fluent)
Spanish - B1 (Basic - Intermediate)
Italian - B1 (Basic - Intermediate)